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Basic Electronics Engineering Workshop

ECE120 · Semester 1 Study Guide

Workshop Instruments ▪ Divider Networks ▪ Diode Physics ▪ Logic Gates ▪ Adders ▪ Arduino Microcontroller

Subject Features:

- Authoritative and comprehensive coverage of all 6 Units of the LPU ECE120 syllabus.
- Passive & active components, resistor color codes, DMM, DSO, and digital trainer module.
- Voltage and current divider laws, potentiometer analysis, and instrument loading corrections.
- Diode V-I characteristics, universal logic gates, half/full adders, and Arduino IR sensor systems.

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CHAPTER 1

Workshop Instruments, Electronic Components, Trainer Module & Display Devices

1.1 Overview and Laboratory Learning Objectives

The foundational practice of electrical and computer engineering requires immediate physical fluency with hardware prototyping platforms, circuit components, precision test instrumentation, and visual indicators. This unit provides the essential laboratory foundation: identifying passive and active discrete components, decoding alphanumeric and color-coded values, operating the bench Digital Multimeter (DMM), Cathode-Ray / Digital Storage Oscilloscope (DSO), Function / Arbitrary Waveform Generator, and the Digital Logic Trainer Module.

Unit Learning Objectives

Upon completing this chapter and accompanying workshop sessions, the engineer will be able to:

1. Differentiate between active and passive electronic components and describe their governing physical mechanisms.
2. Decode 4-band, 5-band, and 6-band carbon and metal-film resistors with tolerance bounds.
3. Read and calculate ceramic, electrolytic, and polyester film capacitor values using EIA 3-digit codes.
4. Measure AC/DC voltages, branch currents, component resistances, diode thresholds, and continuity using a bench DMM.
5. Configure function generators to synthesize sinusoidal, square, and triangular test waveforms with specified DC offsets.
6. Operate a Digital Storage Oscilloscope (DSO) to capture waveforms, calculate timebase period, frequency, and phase difference using Lissajous figures.
7. Safely operate the DC power supplies, logic switches, clock generators, and LED indicators of the digital trainer module.
8. Design current-limiting drive networks for Light Emitting Diodes (LEDs) and wire Common-Cathode / Common-Anode 7-segment displays.

1.2 Classification of Electronic Components

Definition: Passive vs. Active Electronic Components

- **Passive Components**: Devices that do not require an auxiliary energy source to perform their function and cannot generate power gain ($A_P \leq 1$). Examples: Resistors (R), Capacitors (C), Inductors (L), and Transformers.
- **Active Components**: Semiconductor devices capable of controlling electron flow via an external control terminal, producing power amplification, rectification, or switching ($A_P > 1$). Examples: Diodes, Bipolar Junction Transistors (BJTs), Field Effect Transistors (FETs), and Integrated Circuits (ICs).

1.2.1 Resistors and Resistor Color Coding

A resistor dissipates electrical energy as heat according to Joule's Law: $P = I^2R = \frac{V^2}{R}$. Standard through-hole cylindrical resistors indicate resistance value and tolerance via standardized color bands painted circumferentially around the ceramic body.

Resistor Value Derivation

For a 4-band resistor with colors C_1, C_2, C_3, C_4:

$$R = (D_1 \times 10 + D_2) \times 10^{\text{Multiplier}} \pm \text{Tolerance}\,\%$$

For a 5-band precision resistor with colors C_1, C_2, C_3, C_4, C_5:

$$R = (D_1 \times 100 + D_2 \times 10 + D_3) \times 10^{\text{Multiplier}} \pm \text{Tolerance}\,\%$$

Color Band	Digit Value	Multiplier	Tolerance (%)	EIA Band Identifier
Black	0	$10^0 = 1$	—	—
Brown	1	$10^1 = 10$	±1%	F
Red	2	$10^2 = 100$	±2%	G
Orange	3	$10^3 = 1\,\text{k}$	—	—
Yellow	4	$10^4 = 10\,\text{k}$	—	—
Green	5	$10^5 = 100\,\text{k}$	±0.5%	D
Blue	6	$10^6 = 1\,\text{M}$	±0.25%	C
Violet	7	$10^7 = 10\,\text{M}$	±0.1%	B
Gray	8	$10^8 = 100\,\text{M}$	±0.05%	A
White	9	$10^9 = 1\,\text{G}$	—	—
Gold	—	$10^{-1} = 0.1$	±5%	J
Silver	—	$10^{-2} = 0.01$	±10%	K

1.2.2 Capacitors and Numeric Markings

Capacitors store electrical potential in an electrostatic field between conductors: $Q = CV$. Disc ceramic capacitors utilize an EIA 3-digit marking code where the first two digits represent significant figures and the third represents the multiplier in picofarads (pF):

$$C = (D_1 \times 10 + D_2) \times 10^{D_3}\,\text{pF}$$

For instance, a disc capacitor stamped **104K** designates $10 \times 10^4\,\text{pF} = 100{,}000\,\text{pF} = 100\,\text{nF} = 0.1\,\mu\text{F}$ with a $\pm 10\%$ tolerance. Electrolytic capacitors have polarized terminals (longer lead is positive anode; negative cathode indicated by a wide stripe down the package). Reverse biasing electrolytic capacitors causes rapid dielectric breakdown, gas generation, and rupture.

1.3 Laboratory Test and Measurement Instrumentation

1.3.1 The Digital Multimeter (DMM)

A DMM combines a high-precision dual-slope or sigma-delta Analog-to-Digital Converter (ADC) with front-end signal conditioning:

1. **DC/AC Voltage**: High input resistance ($R_{\text{in}} \approx 10\,\text{M}\Omega$) placed in parallel across the device under test.
2. **DC/AC Current**: Low internal shunt resistance ($R_S \approx 0.1$–$1\,\Omega$) placed in series. Current creates a small voltage drop across the shunt ($V_S = IR_S$).
3. **Resistance (Ω)**: The meter injects a constant reference current I_{ref} and measures the resulting voltage drop: $R_x = V/I_{\text{ref}}$.
4. **Diode Test Mode**: Injects a calibrated 1 mA constant current and displays the forward junction potential in volts.

Exam Strategy: DMM Current Mode Danger

Never place a multimeter set to Current mode in parallel across a power source or circuit component. The ammeter shunt acts as a near-dead short ($< 0.1\,\Omega$). Connecting it across +5V will pull tens of amperes, instantly vaporizing the internal fast-blow fuse!

1.3.2 Digital Storage Oscilloscope (DSO)

A DSO captures time-varying waveforms $v(t)$, digitizing voltages through an internal flash ADC and storing them in acquisition memory:

- **Vertical Sensitivity (Volts/Div)**: Sets the amplitude scaling per screen grid square.
- **Horizontal Timebase (Time/Div)**: Sets the duration represented per horizontal grid block.
- **Trigger Controls**: Synchronizes the sweep to a stable repeating edge (rising/falling) preventing display drift.
- **Lissajous Patterns**: When Channel 1 (X) and Channel 2 (Y) receive sinusoidal signals of identical frequency, the resulting ellipse reveals the phase difference ϕ:

$$\sin\phi = \frac{Y_{\text{intercept}}}{Y_{\text{max}}} \implies \phi = \arcsin\left(\frac{Y_1}{Y_2}\right)$$

1.4 Digital Logic Trainer Module Architecture

The bench Digital Logic Trainer Module provides a self-contained environment:

- **Regulated Power Supply Rails**: Fixed +5.0 V DC (≤ 1.0 A) for standard TTL/CMOS circuits, and variable ± 12 V supplies.

- **Logic Input Switches**: SPDT slide or toggle switches producing verified logic LOW (0 V) and HIGH (+5 V).
- **Logic Output LED Monitors**: Buffered high-impedance LED indicators turning ON for logic HIGH (≥ 2.4 V) and OFF for logic LOW (≤ 0.4 V).
- **Internal Pulse Clock Generator**: Selectable clock frequencies (1 Hz, 10 Hz, 1 kHz, 100 kHz) and de-bounced single-pulse pushbuttons for sequential logic testing.
- **Solderless Breadboard Workspace**: Interconnected metal spring clips permitting rapid patch-wire connections.

1.5 Display Devices: LEDs and 7-Segment Displays

1.5.1 Light Emitting Diode (LED) Circuit Design

An LED emits photons through spontaneous radiative recombination across a direct bandgap. Forward voltage drops vary by material:

- Red / Amber LEDs (GaAsP): $V_F \approx 1.8$–2.0 V.
- Green LEDs (GaP): $V_F \approx 2.1$–2.4 V.
- Blue / White LEDs (InGaN): $V_F \approx 3.0$–3.3 V.

To sustain a safe nominal operating current ($I_F \approx 10$–20 mA) from a +5 V supply:

$$R_{\text{limit}} = \frac{V_{CC} - V_F}{I_F} = \frac{5.0\,\text{V} - 2.0\,\text{V}}{15\,\text{mA}} = \frac{3.0}{0.015} = 200\,\Omega \quad \Longrightarrow \text{ Standard } 220\,\Omega$$

1.5.2 Seven-Segment Displays

Seven elongated LEDs (a through g) plus a decimal point (dp) form digits 0 through 9:

- **Common Anode (CA)**: All LED anodes are tied to +5 V. An individual segment lights when its cathode is driven to logic LOW (0 V).
- **Common Cathode (CC)**: All LED cathodes are tied to Ground (0 V). An individual segment lights when its anode is driven to logic HIGH (+5 V).

1.6 Laboratory Experiment 1: Component Identification & Instrument Setup

Definition: Laboratory Experiment 1 Protocol

Objective: To identify passive and active components, decode resistor color codes, verify values with a DMM, and display waveforms on a DSO.
Apparatus Required:

1. Carbon-film resistors ($100\,\Omega, 1\,\text{k}\Omega, 10\,\text{k}\Omega, 100\,\text{k}\Omega$) — 1 each
2. Ceramic disc capacitors ($103, 104, 473$) — 1 each
3. Digital Multimeter (DMM) with test probes — 1 unit
4. Dual-Channel Digital Storage Oscilloscope (DSO) with $10\times$ probes — 1 unit
5. Function Generator — 1 unit
6. Solderless breadboard and patch cords

Observation Table 1.1: Resistor Color Code Verification

S. No.	Band Colors	Nominal Value (R)	Tolerance	Allowed Range	DMM M
1	Brown-Black-Brown-Gold	$100\,\Omega$	$\pm5\%$	95–$105\,\Omega$	99.2
2	Brown-Black-Red-Gold	$1.0\,\text{k}\Omega$	$\pm5\%$	950–$1050\,\Omega$	0.988
3	Brown-Black-Orange-Gold	$10.0\,\text{k}\Omega$	$\pm5\%$	9.5–$10.5\,\text{k}\Omega$	10.04
4	Brown-Black-Yellow-Gold	$100.0\,\text{k}\Omega$	$\pm5\%$	95–$105\,\text{k}\Omega$	98.6

1.7 Worked Numerical Problems

Worked Example: Calculating Resistor Tolerance Bounds

A resistor displays the bands: **Red, Red, Orange, Silver**. Calculate: (a) Nominal value, (b) Upper tolerance limit, (c) Lower tolerance limit.

Step-by-Step Analytical Solution

- Band 1: Red $\implies D_1 = 2$.
- Band 2: Red $\implies D_2 = 2$.
- Band 3: Orange $\implies$ Multiplier $= 10^3 = 1000$.
- Band 4: Silver $\implies$ Tolerance $= \pm10\%$.

Nominal value: $R = 22 \times 10^3\,\Omega = 22\,\text{k}\Omega$.
Tolerance magnitude: $\Delta R = 22\,\text{k}\Omega \times 0.10 = 2.2\,\text{k}\Omega$.
Upper limit: $R_{\max} = 22 + 2.2 = 24.2\,\text{k}\Omega$.
Lower limit: $R_{\min} = 22 - 2.2 = 19.8\,\text{k}\Omega$.

1.8 Workshop Viva-Voce Questions & Answers

1. **Question**: Why does measuring a resistor while in a powered circuit yield inaccurate readings?
   Answer: The DMM resistance circuit operates by injecting a tiny constant current. External operating voltages in an energized circuit distort this current sensing. Furthermore, parallel circuit paths offer alternate current loops, reducing measured equivalent resistance below the true value.
2. **Question**: What is the role of a 10× attenuator probe on an oscilloscope?
   Answer: A 10× passive probe increases the input impedance from 1 MΩ to 10 MΩ and divides the input voltage by 10. This dramatically reduces capacitive and resistive loading on high-frequency and sensitive test points.

CHAPTER 2

Fundamental Circuit Laws & Divider Networks in Workshop

2.1 Overview and Laboratory Learning Objectives

The Voltage Divider Law (VDL) and Current Divider Law (CDL) constitute the two foundational analytical tools used by electrical and computer engineers to determine potential distributions and current branch splits in linear resistive networks. In the workshop environment, understanding these laws is crucial not only for biasing electronic circuits and attenuating sensor signals, but also for evaluating the severe loading errors that arise when measuring instruments or real-world loads are connected across non-ideal dividers.

Unit Learning Objectives

Upon completion of this chapter and related laboratory experiments, the student will be able to:

1. State and mathematically derive the Voltage Divider Law for arbitrary series resistive chains.
2. Explain the operational mechanics of potentiometers and rheostats as adjustable voltage dividers.
3. Quantify voltmeter loading errors mathematically and select appropriate meter impedances to minimize measurement distortion.
4. State and derive the Current Divider Law for two-branch and general n-branch parallel circuits using conductance formulation.
5. Calculate ammeter insertion burden voltage and describe its impact on low-resistance circuits.
6. Build, measure, and troubleshoot series-parallel divider circuits on a solderless breadboard.

2.2 The Voltage Divider Law (VDL)

Definition: Voltage Divider Law

When n resistors are connected in series across a total DC potential difference V_{in}, the current flowing through each element is identical:

$$I = \frac{V_{\text{in}}}{R_{\text{total}}} = \frac{V_{\text{in}}}{\sum_{k=1}^{n} R_k}$$

According to Ohm's Law, the voltage drop across any individual component R_x is directly proportional to its resistance:

$$V_x = I \cdot R_x = V_{\text{in}} \left(\frac{R_x}{R_{\text{total}}} \right)$$

For the standard two-resistor divider formed by R_1 and R_2 across supply V_{in}, the output voltage across R_2 is:

Two-Resistor Voltage Divider Formula

$$V_{\text{out}} = V_{\text{in}} \left(\frac{R_2}{R_1 + R_2} \right)$$

2.2.1 Potentiometer as an Adjustable Voltage Divider

A potentiometer consists of a three-terminal resistive element where a mechanical wiper slides across a track of total resistance R_{pot}. Let the position of the wiper divide the track into a lower portion $R_2 = \alpha R_{\text{pot}}$ and an upper portion $R_1 = (1 - \alpha) R_{\text{pot}}$, where $\alpha \in [0, 1]$ represents the fractional wiper position. The unloaded wiper potential is:

$$V_{\text{out}} = V_{\text{in}} \left(\frac{\alpha R_{\text{pot}}}{(1 - \alpha) R_{\text{pot}} + \alpha R_{\text{pot}}} \right) = \alpha V_{\text{in}}$$

Hence, turning the potentiometer shaft smoothly varies the output voltage from $0\,\text{V}$ to V_{in}.

2.3 Voltmeter Loading Effects and Loading Error

When an external load resistor R_L (or an analog/digital voltmeter with finite internal resistance R_m) is placed across R_2, the equivalent parallel resistance becomes:

$$R_{2,\text{eq}} = R_2 \parallel R_L = \frac{R_2 R_L}{R_2 + R_L}$$

Since $R_{2,\text{eq}} < R_2$, the loaded output voltage drops:

$$V_{\text{loaded}} = V_{\text{in}} \left(\frac{R_{2,\text{eq}}}{R_1 + R_{2,\text{eq}}} \right) = V_{\text{in}} \left(\frac{R_2 R_L}{R_1 R_2 + R_1 R_L + R_2 R_L} \right)$$

Percentage Voltmeter Loading Error

$$\%\ \text{Error} = \frac{V_{\text{unloaded}} - V_{\text{loaded}}}{V_{\text{unloaded}}} \times 100\% = \frac{R_1}{R_1 + R_L \left(1 + \frac{R_1}{R_2}\right)} \times 100\%$$

If $R_L \gg R_2$ (at least 100 times larger), $R_{2,\text{eq}} \approx R_2$ and the loading error drops below 1%.

2.4 The Current Divider Law (CDL)

Definition: Current Divider Law

When n parallel branches are connected across an incoming node current I_T, the voltage across each parallel branch is identical ($V_p = I_T R_p$). The current through any branch k of conductance $G_k = 1/R_k$ is:

$$I_k = I_T \left(\frac{G_k}{\sum_{j=1}^{n} G_j} \right) = I_T \left(\frac{R_p}{R_k} \right)$$

For a two-branch parallel network consisting of R_1 and R_2:

$$R_p = \frac{R_1 R_2}{R_1 + R_2}$$

Substituting into Ohm's Law yields the classical two-branch formulas:

Two-Branch Current Divider Equations

$$I_1 = I_T \left(\frac{R_2}{R_1 + R_2} \right), \quad I_2 = I_T \left(\frac{R_1}{R_1 + R_2} \right)$$

Notice the inverse proportionality: the branch with the smaller resistance carries the larger share of the total current.

2.5 Laboratory Experiment 2: Validation of Voltage and Current Dividers

Laboratory Experiment 2 Protocol

Objective: To experimentally verify the Voltage Divider Law, quantify loading effects with varying load resistors, and verify the Current Divider Law in a parallel branch network.
Apparatus Required:

1. Regulated DC Power Supply (0–30 V) — 1 unit
2. Bench Digital Multimeter (DMM) — 2 units
3. Resistors: $R_1 = 1\,\text{k}\Omega$, $R_2 = 2.2\,\text{k}\Omega$, $R_L = 100\,\Omega, 1\,\text{k}\Omega, 10\,\text{k}\Omega, 100\,\text{k}\Omega$
4. Solderless breadboard and connection jumpers

Observation Table 2.1: Voltage Divider Loading Data ($V_{\text{in}} = 10.0\,\text{V}, R_1 = 1\,\text{k}\Omega, R_2 = 2.2\,\text{k}\Omega$)

Trial	Load Resistor (R_L)	Theor. V_{loaded} (V)	Meas. V_{loaded} (V)	Loading Error (%)	Oper
1	Open Circuit (∞)	6.875 V	6.868 V	0.10%	Ic
2	100 kΩ	6.745 V	6.738 V	1.99%	L
3	10 kΩ	5.909 V	5.902 V	14.07%	Moc
4	1 kΩ	3.571 V	3.565 V	48.14%	Se
5	100 Ω	0.803 V	0.798 V	88.38%	Near

2.6 Worked Numerical Examples

Worked Example: Sizing a Sensor Bias Divider

An embedded analog sensor requiring a stable 3.3 V reference is to be supplied from a +5.0 V Arduino rail using a two-resistor voltage divider. If the maximum current drawn through the divider chain must not exceed 5 mA to save power, calculate: (a) Suitable standard 5% resistor values for R_1 and R_2, (b) The actual voltage produced by these standard values.

Step-by-Step Analytical Solution

Total resistance must satisfy:

$$R_{\text{total}} = R_1 + R_2 \geq \frac{5.0\,\text{V}}{5\,\text{mA}} = 1000\,\Omega$$

Using the divider relation:

$$\frac{R_2}{R_1 + R_2} = \frac{3.3}{5.0} = 0.66 \implies R_2 = 0.66\,R_{\text{total}}, \quad R_1 = 0.34\,R_{\text{total}}$$

Choosing $R_{\text{total}} \approx 3\,\text{k}\Omega$:

$$R_2 \approx 0.66 \times 3000 = 1980\,\Omega \implies \text{Standard } 2.0\,\text{k}\Omega \text{ or } 2.2\,\text{k}\Omega$$

$$R_1 \approx 0.34 \times 3000 = 1020\,\Omega \implies \text{Standard } 1.0\,\text{k}\Omega$$

Testing standard pair $R_1 = 1.0\,\text{k}\Omega$ and $R_2 = 2.0\,\text{k}\Omega$:

$$V_{\text{out}} = 5.0 \times \frac{2.0}{1.0 + 2.0} = 5.0 \times \frac{2}{3} = 3.33\,\text{V}$$

This yields an error of only +1.0%, well within sensor tolerances.

2.7 Workshop Viva-Voce Questions & Answers

1. **Question**: What is the fundamental physical reason why ammeters must have very low internal resistance?
   Answer: An ammeter is connected in series with the branch whose current is being measured. If the meter had non-negligible resistance R_A, it would increase total circuit loop resistance, creating an insertion burden voltage ($V_A = IR_A$) that reduces the true circuit current.
2. **Question**: How does an operational amplifier buffer prevent loading errors in voltage dividers?
   Answer: An op-amp connected as a unity-gain buffer has nearly infinite input impedance ($> 10^{12}\,\Omega$) and extremely low output impedance ($< 0.1\,\Omega$). When placed between the divider output and the load, it draws virtually zero current from the divider (preventing voltage sag) while delivering required load current from its own power supply rails.

CHAPTER 3

Semiconductor PN Junction Diode Characteristics & Workshop Applications

3.1 Overview and Laboratory Learning Objectives

The PN junction diode is the foundational two-terminal non-linear semiconductor building block of modern electronics. Mastering its forward and reverse conduction physics, knee (cut-in) potential, dynamic resistance, and rectifying behavior is required for power supply construction, signal demodulation, voltage clamping, and over-voltage protection circuits.

Unit Learning Objectives

Upon completion of this chapter and lab experiments, the student will be proficient in:

1. Describing the physical mechanism of depletion region formation, built-in barrier potential, and charge carriers.
2. Applying the Shockley diode equation to calculate forward current and thermal voltage V_T.
3. Distinguishing between static (DC) resistance R_D and dynamic (AC) incremental resistance r_d.
4. Performing bench experiments to record forward V-I data points and plot the exponential characteristic curve.
5. Analyzing rectifier topologies (Half-Wave, Full-Wave Center-Tapped, Full-Wave Bridge) and calculating ripple factor, PIV, and efficiency.
6. Diagnosing open, shorted, and leaking diodes on the workbench using DMM diode check mode.

3.2 Physics of the PN Junction

A PN junction is formed by metallurgically joining P-type semiconductor (rich in mobile hole majorities) and N-type semiconductor (rich in electron majorities). At the metallurgical interface:

1. **Carrier Diffusion**: Holes diffuse from P to N, and electrons diffuse from N to P.
2. **Depletion Region**: The migrating carriers recombine, unmasking fixed immobile ionized donor ions (N_D^+) on the N-side and immobile ionized acceptor ions (N_A^-) on the P-side.
3. **Built-in Barrier Potential**: The uncompensated space-charge region establishes an internal electric field $\vec{E}$ pointing from N to P, creating a built-in barrier potential:

$$V_0 = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

At 300 K, $V_0 \approx 0.7\,\text{V}$ for Silicon and $\approx 0.3\,\text{V}$ for Germanium.

3.3 The Shockley Diode Equation

Shockley Diode Characteristic Equation

$$I_D = I_s \left(e^{\frac{V_D}{\eta V_T}} - 1 \right)$$

where:

- I_s is the reverse saturation current (10^{-14} to $10^{-9}\,\text{A}$ for Silicon).
- η is the emission/ideality coefficient ($\eta = 1$ for standard Silicon conduction).
- $V_T = \frac{kT}{q} = \frac{T}{11600} \approx 25.86\,\text{mV} \approx 26\,\text{mV}$ at room temperature (300 K).

3.3.1 Static vs. Dynamic Resistance

Definition: Static and Dynamic Diode Resistance

- **Static (DC) Resistance (R_D)**: The ratio of total DC voltage across the diode to DC current through it at a given quiescent operating point:

$$R_D = \frac{V_D}{I_D}$$

- **Dynamic (AC) Resistance (r_d)**: The reciprocal of the instantaneous slope of the V-I curve at the Q-point:

$$r_d = \frac{dV_D}{dI_D} = \frac{\eta V_T}{I_D + I_s} \approx \frac{\eta V_T}{I_D} = \frac{26\,\text{mV}}{I_D} \quad (\text{at } 300\,\text{K}, \eta = 1)$$

3.4 Laboratory Experiment 3: V-I Characteristics of PN Diode

Laboratory Experiment 3 Protocol

Objective: To experimentally plot the forward-bias V-I characteristic of a Silicon 1N4007 diode and determine its cut-in knee voltage and dynamic resistance.

Apparatus Required:

1. Variable DC Power Supply (0–15 V) — 1 unit
2. 1N4007 Silicon PN junction diode — 1 piece
3. Carbon-film current limiting resistor ($R_S = 1\,\text{k}\Omega, 0.5\,\text{W}$) — 1 piece
4. Digital Multimeters (one for V_D, one for I_D) — 2 units
5. Breadboard and patch jumpers

Observation Table 3.1: Forward V-I Data Points for 1N4007 Silicon Diode

S. No.	Supply Voltage V_S (V)	Diode Drop V_D (V)	Current I_D (mA)	Static R_D (Ω)	Dy
1	0.20	0.200	0.00	∞	
2	0.40	0.395	0.01	39,500	
3	0.60	0.540	0.06	9,000	
4	0.80	0.610	0.19	3,210	
5	1.00	0.645	0.35	1,843	
6	2.00	0.695	1.30	535	
7	4.00	0.725	3.27	222	
8	6.00	0.742	5.26	141	
9	8.00	0.755	7.24	104	
10	10.00	0.765	9.23	83	

3.5 Workshop Rectifier Circuits and Power Conversion

Performance Parameter	Half-Wave	Full-Wave Center-Tapped	Full-Wave Bridge
Diodes Required	1	2	4
DC Output Voltage V_{dc}	$\frac{V_m}{\pi} \approx 0.318\,V_m$	$\frac{2V_m}{\pi} \approx 0.636\,V_m$	$\frac{2V_m}{\pi} \approx 0.636\,V_m$
Ripple Factor γ	1.21	0.482	0.482
Peak Inverse Voltage (PIV)	V_m	$2V_m$	V_m
Rectification Efficiency η	40.6%	81.2%	81.2%

3.6 Worked Numerical Examples

Worked Example: Dynamic Resistance Calculation at Specified Operating Current

A Silicon diode operates with forward current $I_D = 13\,\text{mA}$ at $T = 300\,\text{K}$ with $\eta = 1$. Calculate: (a) Dynamic resistance r_d, (b) AC voltage fluctuation if a small alternating ripple current $\Delta I = \pm 0.5\,\text{mA}$ is superimposed.

Step-by-Step Analytical Solution

(a) Using the dynamic resistance formula:

$$r_d = \frac{\eta V_T}{I_D} = \frac{0.026\,\text{V}}{0.013\,\text{A}} = 2.0\,\Omega$$

(b) The small-signal AC voltage fluctuation across the forward-biased junction is:

$$\Delta V_D = r_d \cdot \Delta I = 2.0\,\Omega \times (\pm 0.5 \times 10^{-3}\,\text{A}) = \pm 1.0\,\text{mV}$$

3.7 Workshop Viva-Voce Questions & Answers

1. **Question**: What causes the reverse breakdown in a Zener diode versus an ordinary rectifier diode?
   Answer: Zener breakdown occurs in heavily doped junctions with narrow depletion regions where intense electric fields ($> 10^7\,\mathrm{V/m}$) rupture covalent bonds by quantum tunneling. Avalanche breakdown occurs in lightly doped junctions where high reverse voltages accelerate minority carriers to kinetic energies sufficient to knock out valence electrons via impact ionization.
2. **Question**: Why does a bridge rectifier require half the PIV rating of a center-tapped rectifier for the same DC output?
   Answer: In a center-tapped full-wave rectifier, during the reverse half-cycle, the non-conducting diode experiences the full end-to-end secondary voltage across both windings ($2V_m$). In a bridge rectifier, the reverse voltage divides across two series reverse-biased diodes, limiting the stress on each diode to V_m.

CHAPTER 4

Digital Logic Gates & Universal Gate Circuit Implementations

4.1 Overview and Laboratory Learning Objectives

Binary logic operations form the mathematical infrastructure of contemporary digital processing, programmable logic, and computer architectures. In this laboratory unit, students verify truth tables of elementary logic gates (NOT, AND, OR, NAND, NOR, XOR, XNOR), understand Transistor-Transistor Logic (TTL) and CMOS electrical thresholds, and apply De Morgan's theorems to synthesize complex Boolean equations exclusively using universal logic gates.

Unit Learning Objectives

Upon completion of this chapter and lab experiments, the student will be proficient in:

1. Defining binary logic levels, noise margins, and standard TTL voltage thresholds ($V_{IH}, V_{IL}, V_{OH}, V_{OL}$).
2. Constructing and validating truth tables for standard 74-series logic gate integrated circuits.
3. Proving algebraically and experimentally that NAND and NOR gates are universal building blocks.
4. Synthesizing NOT, AND, OR, XOR, and XNOR operations using minimum configurations of NAND and NOR gates.
5. Wiring digital ICs safely on a trainer module with proper +5 V VCC, ground, and decoupling capacitors.
6. Troubleshooting common digital workshop errors, including floating high-impedance inputs and inverted chip pinouts.

4.2 Digital Logic Levels and TTL Specifications

Digital electronic systems encode binary information into discrete voltage bands. For the standard TTL 74LS series operating from a $+5.0\,\text{V} \pm 0.25\,\text{V}$ supply:

Logic Parameter	Symbol	Standard TTL Limit
Maximum Output Voltage for Logic LOW	$V_{OL,\max}$	0.4 V
Minimum Output Voltage for Logic HIGH	$V_{OH,\min}$	2.4 V (typical 3.4 V)
Maximum Input Voltage recognized as LOW	$V_{IL,\max}$	0.8 V
Minimum Input Voltage recognized as HIGH	$V_{IH,\min}$	2.0 V
High-Level Noise Margin	$NM_H = V_{OH,\min} - V_{IH,\min}$	$2.4 - 2.0 = 0.4$ V
Low-Level Noise Margin	$NM_L = V_{IL,\max} - V_{OL,\max}$	$0.8 - 0.4 = 0.4$ V

The intermediate region between 0.8 V and 2.0 V is the indeterminate forbidden state. Floating TTL inputs naturally drift into this region, acting as antennas for high-frequency electromagnetic noise.

4.3 Standard 74-Series Integrated Circuit Architectures

Definition: 74-Series IC Pinout Mapping

Standard 74-series TTL logic gates are packaged in 14-pin Dual In-line Packages (DIP). In standard convention, **Pin 7** connects to Circuit Ground (0 V) and **Pin 14** connects to +5.0 V (V_{CC}):

- **7404 Hex Inverter**: Six independent NOT gates. Pin pairs: (1 in, 2 out), (3 in, 4 out), (5 in, 6 out), (9 in, 8 out), (11 in, 10 out), (13 in, 12 out).
- **7408 Quad 2-Input AND**: Four independent AND gates. Pin pairs: (1,2 in, 3 out), (4,5 in, 6 out), (9,10 in, 8 out), (12,13 in, 11 out).
- **7432 Quad 2-Input OR**: Four independent OR gates. Same pin mapping as 7408.
- **7400 Quad 2-Input NAND**: Four independent NAND gates. Same pin mapping as 7408.
- **7402 Quad 2-Input NOR**: Four independent NOR gates. **Caution**: Non-standard inverted pinout! Output on Pin 1, inputs on Pins 2 and 3; output on Pin 4, inputs on Pins 5 and 6!
- **7486 Quad 2-Input XOR**: Four independent Exclusive-OR gates ($Y = A \oplus B = \overline{A}B + A\overline{B}$).

4.4 Universal Gate Syntheses (NAND and NOR)

A logic gate family is classified as *universal* if any arbitrary combinational Boolean function can be realized exclusively using that gate type without requiring other gate families.

Theorem: De Morgan's Theorems

$$\overline{A \cdot B} = \overline{A} + \overline{B}$$
$$\overline{A + B} = \overline{A} \cdot \overline{B}$$

4.4.1 Realizations Using Only NAND Gates

1. **NOT Gate** ($\overline{A}$): Tie both inputs together:

$$Y = \overline{A \cdot A} = \overline{A} \quad (1 \text{ Gate})$$

2. **AND Gate** ($A \cdot B$): Pass NAND output through a NAND inverter:

$$Y = \overline{\overline{A \cdot B} \cdot \overline{A \cdot B}} = A \cdot B \quad (2 \text{ Gates})$$

3. **OR Gate** ($A + B$): Invert inputs individually and feed into a third NAND gate:

$$Y = \overline{\overline{A} \cdot \overline{B}} = \overline{\overline{A}} + \overline{\overline{B}} = A + B \quad (3 \text{ Gates})$$

4. **XOR Gate** ($A \oplus B$): Realized using 4 NAND gates:

$$Y = \overline{\overline{A \cdot \overline{AB}} \cdot \overline{B \cdot \overline{AB}}} = A\overline{B} + \overline{A}B \quad (4 \text{ Gates})$$

4.5 Laboratory Experiment 4: Universal Gate Implementations

Laboratory Experiment 4 Protocol

Objective: To wire and experimentally verify the truth tables of NOT, AND, OR, and XOR gates synthesized exclusively from a 7400 Quad 2-Input NAND IC.
Apparatus Required:

1. Digital Logic Trainer Module with regulated +5 V supply — 1 unit
2. IC 7400 Quad 2-Input NAND gate — 1 piece
3. Logic input toggle switches and logic monitor LEDs
4. Patch wires and breadboard

Observation Table 4.1: Truth Table Verification of NAND-Synthesized Gates

Input A	Input B	NOT ($\overline{A}$)	AND ($A \cdot B$)	OR ($A + B$)	XOR ($A \oplus B$)	Trainer Indicator
0	0	1	0	0	0	Matches Theor
0	1	1	0	1	1	Matches Theor
1	0	0	0	1	1	Matches Theor
1	1	0	1	1	0	Matches Theor

4.6 Worked Logic Design Problems

Worked Example: Simplifying and Implementing a Logic Function with NAND Gates

Given the Boolean expression $F = AB + \overline{A}C$, synthesize this function using the minimum number of 2-input NAND gates.

Step-by-Step Analytical Solution

Apply double negation to express F in NAND-NAND form:

$$F = \overline{\overline{AB + \overline{A}C}}$$

Applying De Morgan's theorem to the outer negation:

$$F = \overline{\overline{AB} \cdot \overline{\overline{A}C}}$$

Gate implementation:

- Gate 1 (Inverter): Invert A to yield $\overline{A}$ using 1 NAND gate: $\overline{A} = \overline{A \cdot A}$.
- Gate 2: Compute $G_1 = \overline{AB}$ using 1 NAND gate.
- Gate 3: Compute $G_2 = \overline{\overline{A}C}$ using 1 NAND gate.
- Gate 4: Feed G_1 and G_2 into a final NAND gate: $F = \overline{G_1 \cdot G_2}$.

Total gate count: exactly **4 NAND gates** (one single 7400 IC!).

4.7 Workshop Viva-Voce Questions & Answers

1. **Question**: Why is a 0.1 μF ceramic capacitor placed between Pin 14 (V_{CC}) and Pin 7 (GND) of logic ICs on a breadboard?
   Answer: When internal TTL or CMOS transistors switch states, they briefly short power to ground for a few nanoseconds, drawing current spikes (di/dt). Due to breadboard lead inductance, this creates high-frequency voltage glitches on the power rail. The decoupling capacitor acts as a localized charge reservoir, shunting high-frequency noise directly to ground.
2. **Question**: What happens if you leave unused inputs floating on a CMOS 74HC00 IC?
   Answer: CMOS inputs have virtually infinite gate resistance ($> 10^{12}\,\Omega$). A floating input will float unpredictably into the linear threshold region between V_{IL} and V_{IH}, turning both the PMOS and NMOS internal transistors partially ON simultaneously. This causes excessive shoot-through current, overheating the IC and causing thermal destruction.

CHAPTER 5

Combinational Arithmetic & Logic Circuits: Half/Full Adders & Subtractors

5.1 Overview and Laboratory Learning Objectives

Arithmetic Logic Units (ALUs) form the computational core of microprocessors, digital signal processors, and graphics pipelines. The most elementary mathematical operation implemented in hardware is binary addition. This chapter explores the systematic gate-level design, truth-table derivation, hardware breadboarding, and diagnostic verification of Half Adders, Full Adders, Half/Full Subtractors, and 4-bit parallel binary adders.

Unit Learning Objectives

Upon completion of this chapter and lab experiments, the student will be proficient in:

1. Formulating truth tables for binary addition and binary subtraction.
2. Deriving minimal sum-of-products Boolean expressions for Sum, Carry, Difference, and Borrow.
3. Designing and wiring a Half Adder using discrete XOR and AND gates.
4. Designing and wiring a Full Adder by cascading two Half Adders and an OR gate.
5. Implementing arithmetic adders using universal NAND gates exclusively.
6. Analyzing the operation and carry propagation delay of the IC 7483 4-bit binary adder.
7. Troubleshooting combinational glitches and wiring errors on digital trainer modules.

5.2 Binary Addition Principles and the Half Adder

Binary addition follows four primitive single-bit rules:

$$0+0=0, \quad 0+1=1, \quad 1+0=1, \quad 1+1=10_2 \text{ (Sum 0 with Carry 1)}$$

Definition: Half Adder

A Half Adder is a combinational circuit with two single-bit inputs (A, B) and two outputs: Sum (S) and Carry (C).

Input A	Input B	Sum (S)	Carry (C)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Extracting Boolean minterms directly from the truth table:

Half Adder Boolean Expressions

$$S = \overline{A}B + A\overline{B} = A \oplus B$$
$$C = A \cdot B$$

Hardware implementation requires:

- One 2-input XOR gate (1/4th of IC 7486) for Sum.
- One 2-input AND gate (1/4th of IC 7408) for Carry.

5.3 The Full Adder Architecture

When multi-bit binary numbers are added, each column except the least significant bit must accommodate an incoming carry (C_{in}) generated by the preceding stage.

Definition: Full Adder

A Full Adder is a combinational circuit with three single-bit inputs: A, B, and C_{in}, and two outputs: Sum (S) and Carry Out (C_{out}).

Input A	Input B	Input C_{in}	Sum (S)	Carry Out (C_{out})
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

5.3.1 Boolean Derivation for Full Adder

The Sum output minterms are:

$$S = \sum m(1, 2, 4, 7) = \overline{A}\overline{B}C_{\text{in}} + \overline{A}B\overline{C_{\text{in}}} + A\overline{B}\overline{C_{\text{in}}} + ABC_{\text{in}}$$

Factoring out C_{in}:

$$S = C_{\text{in}}(\overline{A}\overline{B} + AB) + \overline{C_{\text{in}}}(\overline{A}B + A\overline{B}) = C_{\text{in}}\overline{(A \oplus B)} + \overline{C_{\text{in}}}(A \oplus B) = A \oplus B \oplus C_{\text{in}}$$

The Carry Out minterms are:

$$C_{\text{out}} = \sum m(3,5,6,7) = AB + BC_{\text{in}} + AC_{\text{in}} = AB + C_{\text{in}}(A+B) = AB + C_{\text{in}}(A \oplus B)$$

Full Adder Canonical Expressions

$$S = A \oplus B \oplus C_{\text{in}}$$
$$C_{\text{out}} = AB + C_{\text{in}}(A \oplus B)$$

5.3.2 Implementation Using Two Half Adders and an OR Gate

1. **Half Adder 1**: Inputs $A, B \implies S_1 = A \oplus B$, $C_1 = AB$.
2. **Half Adder 2**: Inputs $S_1, C_{\text{in}} \implies S = S_1 \oplus C_{\text{in}} = (A \oplus B) \oplus C_{\text{in}}$, $C_2 = S_1 \cdot C_{\text{in}} = (A \oplus B)C_{\text{in}}$.
3. **OR Gate**: $C_{\text{out}} = C_1 + C_2 = AB + (A \oplus B)C_{\text{in}}$.

5.4 Laboratory Experiment 5: Breadboarding and Testing Adders

Laboratory Experiment 5 Protocol

Objective: To design, assemble on a solderless breadboard, and functionally verify the complete truth tables of a Half Adder and a Full Adder.
Apparatus Required:

1. Digital Trainer Module with +5 V supply and logic switches — 1 unit
2. IC 7486 (Quad 2-Input XOR) — 1 piece
3. IC 7408 (Quad 2-Input AND) — 1 piece
4. IC 7432 (Quad 2-Input OR) — 1 piece
5. Breadboard and patch jumpers

Observation Table 5.1: Full Adder Truth Table Verification on Trainer

Input A	Input B	Input C_{in}	Observed Sum LED	Observed Carry LED
0	0	0	OFF (0)	OFF (0)
0	0	1	ON (1)	OFF (0)
0	1	0	ON (1)	OFF (0)
0	1	1	OFF (0)	ON (1)
1	0	0	ON (1)	OFF (0)
1	0	1	OFF (0)	ON (1)
1	1	0	OFF (0)	ON (1)
1	1	1	ON (1)	ON (1)

5.5 Worked Numerical Examples

Worked Example: Multi-Bit Binary Addition and Propagation Delay

A 4-bit ripple-carry adder is constructed by cascading four Full Adders. Each Full Adder has a gate delay of 12 ns for Sum and 8 ns for Carry generation. (a) Compute the result of adding $A = 1011_2$ (11) and $B = 0110_2$ (6). (b) Determine the maximum propagation delay required to establish the final valid 4-bit sum and carry-out.

Step-by-Step Analytical Solution

(a) Binary addition calculation:

Carry:	1	1	0	0	
A:		1	0	1	1
B:		0	1	1	0
Sum:	1	0	0	0	1

The result is $10001_2 = 16 + 1 = 17_{10}$, verifying arithmetic correctness.
(b) Propagation Delay: In a ripple carry adder, the carry must ripple through all four stages:

$$t_{\text{carry,total}} = 4 \times t_{\text{carry}} = 4 \times 8\,\text{ns} = 32\,\text{ns}$$

The final MSB sum bit requires the final carry plus the XOR sum delay:

$$t_{\text{total}} = 3 \times t_{\text{carry}} + t_{\text{sum}} = (3 \times 8) + 12 = 24 + 12 = 36\,\text{ns}$$

5.6 Workshop Viva-Voce Questions & Answers

1. **Question**: What is the difference between a ripple-carry adder and a carry-lookahead adder?
   Answer: In a ripple-carry adder, each Full Adder stage must wait for the carry bit to propagate sequentially from preceding stages, resulting in a total delay proportional to n ($O(n)$). A carry-lookahead adder computes carry bits in parallel across all stages simultaneously using Generate ($G_i = A_i B_i$) and Propagate ($P_i = A_i \oplus B_i$) logic, reducing the delay to a constant $O(1)$ independent of bit width.
2. **Question**: How can a binary adder be modified to perform binary subtraction without extra subtractor ICs?
   Answer: By utilizing 2's complement arithmetic ($A - B = A + \overline{B} + 1$). Inputs B are passed through XOR gates controlled by a Mode line (M). When $M = 1$, the XOR gates invert B to produce the 1's complement ($\overline{B}$), and setting the initial carry $C_0 = M = 1$ adds 1, completing the 2's complement subtraction.

CHAPTER 6

Microcontroller Workshop: Arduino Uno & Embedded Sensor Interfacing

6.1 Overview and Laboratory Learning Objectives

The integration of physical electronic hardware with programmable digital controllers represents the core of modern robotics, Internet of Things (IoT), industrial instrumentation, and automated control systems. In this culminating workshop unit, students explore the architecture of the open-source Arduino Uno development platform (powered by the Microchip ATmega328P microcontroller), master C/C++ embedded firmware programming, interface an active Infrared (IR) Proximity/Obstacle sensor, and implement real-world automated control and telemetry algorithms.

Unit Learning Objectives

Upon completion of this chapter and lab experiments, the student will be proficient in:

1. Explaining the hardware architecture, memory map, and peripheral subsystems of the ATmega328P.
2. Mapping physical Arduino Uno header pins to digital I/O ports, PWM channels, and 10-bit analog inputs.
3. Developing structured embedded firmware using standard Arduino IDE sketch architecture (‘setup()‘ and ‘loop()‘).
4. Operating the LM393 comparator-based active Infrared (IR) obstacle detection module.
5. Calibrating sensor detection thresholds using an on-board multi-turn cermet potentiometer.
6. Implementing software debounce and state machine logic to build reliable item counters and intrusion alert systems.
7. Diagnosing electrical hazards such as inductive kickback, brownout resets, and ground bounce.

6.2 Arduino Uno Hardware Architecture

The Arduino Uno R3 is built around the 8-bit AVR RISC ATmega328P microcontroller:

- **Processing Core**: 8-bit AVR RISC architecture executing up to 16 MIPS at 16 MHz.
- **Memory Organization**:
  - 32 KB In-System Programmable Flash memory (0.5 KB reserved for Optiboot bootloader).

  - 2 KB internal SRAM for dynamic variable allocation and stack frame execution.
  - 1 KB EEPROM for persistent non-volatile data storage across power cycles.
- **Digital I/O (Pins 0 to 13)**: 14 general-purpose digital input/output pins. Absolute maximum current per pin is 40 mA (20 mA recommended continuous limit).
- **Pulse Width Modulation (PWM)**: 6 hardware timer channels (Pins 3, 5, 6, 9, 10, 11 marked with ∼) producing 8-bit PWM signals (0–255) at 490 Hz or 980 Hz.
- **Analog Inputs (Pins A0 to A5)**: 6 single-ended channels multiplexed into a 10-bit successive approximation ADC.

$$V_{\text{meas}} = \text{ADC Reading} \times \left(\frac{5.0\,\text{V}}{1024}\right) \approx \text{ADC Reading} \times 4.8828\,\text{mV}$$

- **Power Distribution**: Regulated on-board 5.0 V (NCP1117 LDO) and 3.3 V (LP2985 LDO) outputs from a 7–12 V DC barrel jack input.

6.3 Infrared (IR) Proximity and Obstacle Detection Sensor Module

The standard workshop IR sensor module utilizes optical triangulation and reflection:

1. **IR Emitter (LED)**: Transmits continuous infrared light at a wavelength of $\lambda \approx 940\,\text{nm}$.
2. **IR Detector (Photodiode)**: Measures reflected infrared intensity. When an obstacle approaches, reflected radiation enters the photodiode, generating a proportional reverse photocurrent.
3. **LM393 Dual Differential Comparator**: Compares the photodiode voltage (V_{photo}) against an adjustable reference voltage (V_{ref}) configured via a 10-turn cermet potentiometer:

$$V_{\text{OUT}} = \begin{cases} 0\,\text{V (Logic LOW / Active)}, & \text{when } V_{\text{photo}} > V_{\text{ref}} \text{ (Obstacle detected)} \\ 5\,\text{V (Logic HIGH / Idle)}, & \text{when } V_{\text{photo}} \leq V_{\text{ref}} \text{ (No obstacle)} \end{cases}$$

4. **Status Indicators**: A power LED (red) confirms +5 V supply; an output indicator LED (green) illuminates when an obstacle is detected.

6.4 Laboratory Experiment 6: Arduino IR Obstacle Detection & Counter

Laboratory Experiment 6 Protocol

Objective: To interface an active-LOW IR sensor module and an alert actuator to an Arduino Uno, calibrate detection distance, and write embedded firmware implementing a debounced conveyor counter.

Apparatus Required:

1. Arduino Uno R3 development board with USB cable — 1 unit
2. 3-Pin Active-LOW IR Proximity Sensor Module (LM393) — 1 unit
3. Red LED (5 mm) with 220 Ω resistor — 1 set
4. Active 5 V Piezo Buzzer — 1 unit
5. Solderless breadboard and female-to-male DuPont jumpers

Observation Table 6.1: IR Sensor Calibration and Distance Response

Target Distance (cm)	White Surface (OUT)	Cardboard (OUT)	Black Matte (OUT)	D
2.0	0.0 V (LOW)	0.0 V (LOW)	0.0 V (LOW)	
5.0	0.0 V (LOW)	0.0 V (LOW)	4.8 V (HIGH)	Su
10.0	0.0 V (LOW)	4.8 V (HIGH)	5.0 V (HIGH)	
15.0	4.9 V (HIGH)	5.0 V (HIGH)	5.0 V (HIGH)	
20.0	5.0 V (HIGH)	5.0 V (HIGH)	5.0 V (HIGH)	

6.5 Complete Embedded Firmware Implementation

Production-Ready Arduino C++ Counter Sketch

```
/*
 * ECE120 Basic Electronics Engineering Workshop
 * Arduino Uno & LM393 IR Obstacle Counter with Software Debounce
 */

const int IR_PIN = 2;       // Digital Pin 2 (Active LOW)
const int LED_PIN = 13;     // On-board LED indicator
const int BUZZER_PIN = 8;   // Audible buzzer alert pin

volatile unsigned long objectCount = 0;
unsigned long lastEventTime = 0;
const unsigned long DEBOUNCE_MS = 120; // 120 ms minimum pulse separation

void setup() {
  pinMode(IR_PIN, INPUT_PULLUP);
  pinMode(LED_PIN, OUTPUT);
  pinMode(BUZZER_PIN, OUTPUT);

  digitalWrite(LED_PIN, LOW);
  digitalWrite(BUZZER_PIN, LOW);

  Serial.begin(9600);
  while (!Serial) { ; } // Wait for serial monitor connection
  Serial.println("========================================");
  Serial.println("ECE120: Arduino Uno IR Object Counter");
  Serial.println("System Initialized & Ready");
  Serial.println("========================================");
}

void loop() {
  int sensorVal = digitalRead(IR_PIN);

  // Check for active LOW signal from LM393
```

```
  if (sensorVal == LOW) {
    unsigned long currentMillis = millis();
    if (currentMillis - lastEventTime > DEBOUNCE_MS) {
      objectCount++;
      lastEventTime = currentMillis;

      // Visual feedback
      digitalWrite(LED_PIN, HIGH);

      // Real-time serial telemetry
      Serial.print("[TELEMETRY] Object Detected! Cumulative Count: ");
      Serial.println(objectCount);

      // Trigger buzzer if count reaches batch threshold (e.g. 20 objects)
      if (objectCount % 20 == 0) {
        Serial.println(">>> BATCH THRESHOLD REACHED: AUDIBLE ALARM <<<");
        digitalWrite(BUZZER_PIN, HIGH);
        delay(100);
        digitalWrite(BUZZER_PIN, LOW);
      }

      delay(40);
      digitalWrite(LED_PIN, LOW);
    }
  }
}
```

6.6 Workshop Viva-Voce Questions & Answers

1. **Question**: Why does the microcontroller experience unintended brownout resets when inductive motors or relays are switched?
   Answer: Inductive coils store energy in a magnetic field ($E = \frac{1}{2}LI^2$). When turned OFF suddenly, the collapsing field generates a massive reverse back-EMF spike ($V = -L\frac{di}{dt}$) that pulls down the regulated DC rail. If the supply voltage drops below the ATmega328P internal Brown-Out Detector (BOD) threshold of 4.3 V or 2.7 V, the CPU executes an emergency hardware reset. Placing a reverse-biased flyback diode across the coil shunts this transient harmlessly.
2. **Question**: What is the purpose of software debounce in mechanical and optical counter systems?
   Answer: When an object partially crosses an optical sensor beam or when mechanical switch contacts engage, micro-vibrations and edge reflections cause the signal to rapidly toggle between HIGH and LOW dozens of times over a 5–50 ms window. Software debounce enforces a mandatory lockout hold-off window (e.g. 100 ms) after the initial edge, ensuring only one valid count is recorded per physical object.